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Problem set number: 3

Due date: 2/24/26 (11:59)

Date of submission: 2/24/26

Acknowledgments:

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Question 1:

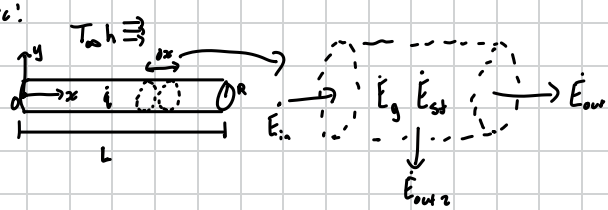
Known: - rod is of radius R and length L and uniform volumetric heat generation of $\dot{q}$
- L) placed in environment of T_∞, h
- (for part (c)) end of rod at $x=0$ is maintained at constant temp T_1

Find: a) the energy balance eqn for control volume, with each energy term in the eqn as a function of T, when considering a differential cylinder with a height of ∂x as the control volume

b) The governing eqn that describes the temperature profile of the rod, simplified as much as possible

c) The boundary and initial conditions for the governing diff eqn when at $t=0$, the rod has uniform initial temp of T_i

Schematic:



Assumptions: 1) HT due to radiation can be neglected
2) radius R is very small → 5) essentially only 1D Heat transfer
3) temperature of rod in radial direction is uniform → $T = T(x,t)$
4) k, ρ, C_p are all constant

Analysis:

a) $\dot{E}_{in} = q_x$ $\quad E_{out,1} = q_{x+\partial x}$ $\quad \dot{E}_{out\,2} = q''_{conv} = h(2\pi R\,\partial x)(T(x,t), T_\infty) = h \cdot 2\pi R(T - T_\infty)\,\partial x$

$\dot{E}_g = \dot{q}\pi R^2 \partial x$ $\quad E_{st} = \rho C_p \pi R^2 \partial x \frac{\partial T}{\partial x}$ $\quad \frac{\partial q_x}{\partial x} = \frac{\partial}{\partial x}\left(-kA\frac{\partial T}{\partial x}\right) = -k\pi R^2 \partial x \frac{\partial^2 T}{\partial x^2}$

$\dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = \dot{E}_{st} \Rightarrow \quad q_x + \dot{q}\pi R^2 \partial x - \left(q_x + \frac{\partial q_x}{\partial x}\right) - h(T_\infty - T) = \rho C_p \pi R^2 \frac{\partial T}{\partial t}\partial x$

$$\Rightarrow \boxed{\underbrace{-k\pi R^2 \frac{\partial T}{\partial x}}_{\dot{E}_{in}} + \underbrace{\dot{q}\pi R^2 \partial x}_{\dot{E}_g} + \underbrace{k\pi R^2 \frac{\partial T}{\partial x} + k\pi R^2 \frac{\partial^2 T}{\partial x^2}\partial x}_{\dot{E}_{out\,1}} - \underbrace{h\,2\pi R(T - T_\infty)\partial x}_{\dot{E}_{out\,2}} = \underbrace{\rho C_p \pi R^2 \frac{\partial T}{\partial t}\partial x}_{\dot{E}_{st}}}$$

b) $$-\cancel{k\pi R^2 \frac{\partial T}{\partial x}} + \dot{q}\cancel{\pi R^2}\cancel{\partial x} + \cancel{k\pi R^2 \frac{\partial T}{\partial x}} + k\cancel{\pi R^2}\frac{\partial^2 T}{\partial x^2}\cancel{\partial x} - h\frac{2\cancel{\pi}\cancel{R}}{R}(T - T_\infty)\cancel{\partial x} = \rho C_p \cancel{\pi R^2}\frac{\partial T}{\partial t}\cancel{\partial x}$$

$$\dot{q} + k\frac{\partial^2 T}{\partial x^2} - \frac{2}{R}h(T - T_\infty) = \rho C_p \frac{\partial T}{\partial t} \Rightarrow \boxed{\rho C_p \frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x} + \dot{q} - \frac{2h}{R}(T - T_\infty)}$$

c) $T(0,t) = T_1$

$T(x,0) = T_i$ thus $\boxed{-k\frac{\partial T}{\partial x}\Big|_{x=L} = h\left(T_\infty - T(L,0)\right)}$

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Question 2:

Known:
- Composite wall made up of walls 1 and 2
- $L_A = 30cm$, and experiences uniform volumetric heat generation $\dot{q}$
- $L_B = 10cm$, $K_B = 30\ W/mK$
- left Surface wall A is insulated, right Surface wall A is in contact with left side wall B, right Side of wall B is exposed to coolant at $T_\infty = 20°C$ with $h = 500\ W/m^2K$
- Sensor at $x=0$ reads $T_w = 115°C$, Sensor at $x = 30cm$ reads $T_i = 100°C$

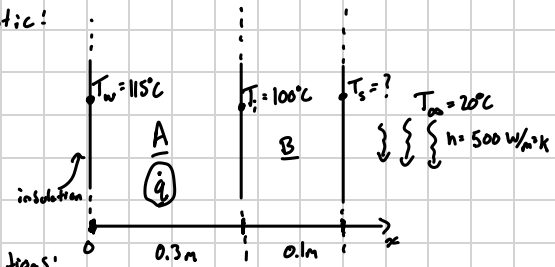
Find:
a) T_s, where T_s = temp of right Surface of wall B $= T(x = 40cm)$

b) $\dot{q}$

c) thermal conductivity of wall A, K_A

d) a plot of temperature profile across the composite wall, with important features shown

Schematic:



Assumptions:
1. 1D heat transfer across composite wall
2. radiation can be ignored
3. No contact thermal resistance at interface between A and B

Analysis:

a) $\dot{q}_{conv} = \dot{q}_{cond,B}$

$$\dot{q}_{conv} = hA(T_s - T_\infty) = -K_B A\left(\frac{T_s - T_i}{L_B}\right) = \dot{q}_B \quad , T_\infty = 20°C + 273 = 293K,\ T_i = 100°C + 273 = 373°C$$

$$500\ W/m^2K\,(T_s - 293K) = -(30\ W/mK)\left(\frac{T_s - 373K}{0.1m}\right)$$

$$T_s = 323K \Rightarrow \boxed{T_s = 50°C}$$

b) $\dot{q} A L_A = \dot{q}_{conv} = hA(T_s - T_\infty)$

$$\dot{q} = \frac{h(T_s - T_\infty)}{L_A} = \frac{500\ W/m^2K\,(323K - 293K)}{0.3m} = 50000\ W/m^3 \qquad \boxed{\dot{q} = 50\ \frac{kW}{m^3}}$$

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Question 2 (cont.)

c) $K_A \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$

$$\frac{\partial^2 T}{\partial x^2} = \frac{-\dot{q}}{K_A} \Rightarrow \int \frac{\partial^2 T}{\partial x^2} \partial x = \int \frac{-\dot{q}}{K_A} \partial x \Rightarrow \frac{\partial T}{\partial x} = \frac{-\dot{q}x}{K_A} + C, \quad \left.\frac{\partial T}{\partial x}\right|_{x=0} = 0 \rightarrow 0 = 0 + C \Rightarrow C = 0$$

$$\Rightarrow \int \frac{\partial T}{\partial x} \partial x = \int -\frac{\dot{q}x}{K_A} \partial x \Rightarrow T(x,\infty) = \frac{-\dot{q}x^2}{2K_A} + D, \quad T(0,\infty) = T_W \rightarrow T_W = 0 + D \Rightarrow D = T_W$$

thus $T(x,\infty) = \frac{-\dot{q}x^2}{2K_A} + T_W$

$$T(L_A,\infty) = T_i = \frac{-\dot{q}T_i^2}{2K_A} + T_W \Rightarrow 373\,K = \frac{-50000\,W/m^3 (0.3m)^2}{2 \cdot K_A} + 388\,K$$

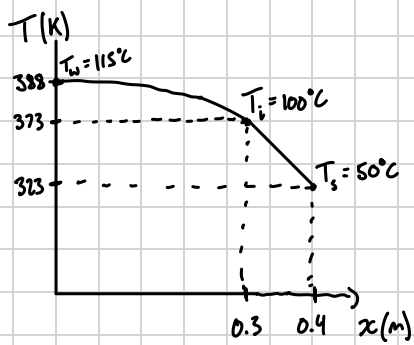
$$30K \cdot K_A = 4500\,W/m \Rightarrow \boxed{K_A = 150\,W/mK}$$

d) $T(x,\infty) = \frac{-\dot{q}x^2}{2K_A} + T_W$

$$\Rightarrow T(x,\infty) = \frac{-50000\,W/m^3 \cdot x^2}{2 \cdot 150\,W/mK} + 388\,K$$

$$\Rightarrow T(x,\infty) = -166.67x^2 \cdot \left(\frac{1K}{m^2}\right) + 388K \quad \underline{\text{for } 0 < x < 0.3}$$

$$T(x,\infty) = -500x \cdot \left(\frac{1K}{m}\right) + 523K \quad \underline{\text{for } 0.3 < x < 0.4}$$



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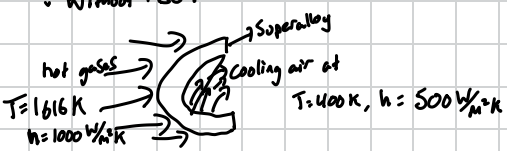
Question 3

Known:
- blade is made with Superalloy, $k_b = 25\ W/mK$
- TBC is made of ceramic material, $k_{TBC} = 1\ W/mK$
- metallic bonding agent creates interfacial thermal resistance (/unit area) of $R'' = 10^{-4}\ m^2K/W$
- In operating condition, external air at $T_{\infty,o} = 1616K$ with $h_o = 1000\ W/m^2K$
  internal cooling air is $T_{\infty,i} = 400K$ with $h_i = 500\ W/m^2K$
- thickness of Superalloy blade is $L_b = 5mm$, thickness of TBC is $0.5\ mm$
- Superalloy $T_{max} = 1200K$

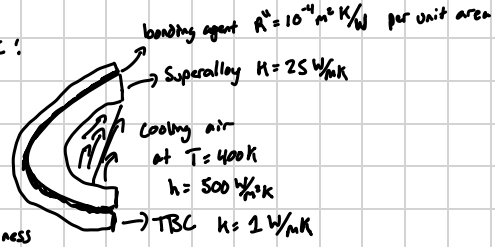
Find:
a) exterior surface temp of Superalloy blade if TBC is not applied.
Will blade be maintained below T_{max}?
b) exterior surface temp of blade when TBC is applied.
Will blade be maintained below T_{max}?

Schematic:

- without TBC:



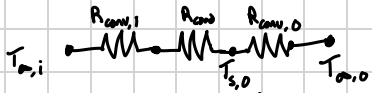
With TBC:



Assumptions:
1. blade at steady state
2. thermal radiation is negligible
3. 1D heat transfer through TBC and Superalloy
4. bonding thickness is negligible

Analysis:

a) drawing of thermal resistance network:



convection resistance $= \frac{1}{hA}$ conduction resistance $= \frac{L}{kA}$

thus $R_{conv,i} = \frac{1}{h_iA} = \frac{1}{500\ W/m^2K \cdot A}$, $R_{conv,o} = \frac{1}{h_oA} = \frac{1}{1000\ W/m^2K \cdot A}$, $R_{cond} = \frac{0.005m}{25\ W/mK \cdot A}$

$$R_{tot} = \frac{2}{1000\ W/m^2K \cdot A} + \frac{1}{1000\ W/m^2K \cdot A} + \frac{0.2m}{1000\ W/mK \cdot A} = \frac{3.2}{1000\ W/m^2K \cdot A} = 0.0032\ \frac{m^2K}{W} \cdot A^{-1}$$

$$q = \frac{T_{\infty,o} - T_{\infty,i}}{R_{tot}} = \frac{1616K - 400K}{0.0032\ \frac{m^2K}{W} \cdot A^{-1}} = 3.8 \times 10^5\ \frac{W}{m^2} \cdot A$$

$$q = \frac{\Delta T_{conv,o}}{R_{conv,o}} \Rightarrow \Delta T_{conv,o} = q \cdot R_{conv,o} = T_{\infty,o} - T_s \Rightarrow T_s = T_{\infty,o} - q R_{conv,o}$$

$$T_s = 1616K - \left(3.8 \times 10^5\ \frac{W}{m^2} \cdot A\right)\left(\frac{1}{1000\ W/m^2K \cdot A}\right)$$

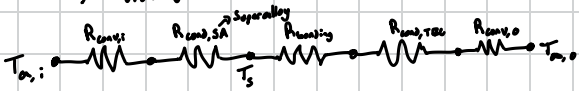
$$\boxed{T_s = 1236\ K}$$

Thus exterior surface of Superalloy cannot be maintained under 1200K, and alloy will get too hot

Question 3 (cont.)

Analysis (cont.):

b) drawing of thermal resistance network:



$$R_{conv,i+cond,SA+conv,o} = \frac{3.2}{1000\ W/m^2K \cdot A} \quad \text{from part (a)}$$

$$R_{cond,TBC} = \frac{0.0005\,m}{1\ W/mK \cdot A} = \frac{0.5}{1000\ W/m^2K \cdot A}$$

$$\text{thus } R_{total} = \frac{3.8}{1000\ W/m^2K \cdot A} = 0.0038\ \frac{m^2K}{W} \cdot A^{-1}$$

$$R_{bonding} = \frac{10\times10^{-4}}{A}\ \frac{m^2K}{W} = \frac{0.1}{1000\ W/m^2K \cdot A}$$

$$q = \frac{T_{\infty,o} - T_{\infty,i}}{R_{total}} = \frac{1616\,K - 400\,K}{0.0038\ \frac{m^2K}{W} \cdot A^{-1}} = 3.2\times10^5\ W/m^2 \cdot A$$

$$q = \frac{\Delta T_{conv,o+cond,TBC+bonding}}{A_{conv,o+cond,TBC+bonding}} \Rightarrow \Delta T_{conv,o+cond,TBC+bonding} = q \cdot A_{conv,o+cond,TBC+bonding} = T_{\infty,o} - T_s$$

$$A_{conv,o+cond,TBC+bonding} = \frac{1}{1000\ W/m^2K \cdot A} + \frac{0.5}{1000\ W/m^2K \cdot A} + \frac{0.1}{1000\ W/m^2K \cdot A} = \frac{1.6}{1000\ W/m^2K \cdot A} = 0.0016\ \frac{m^2K}{W} \cdot A^{-1}$$

$$T_s = T_{\infty,o} - q \cdot A_{conv,o+cond,TBC+bonding} = 1616\,K - (3.2\times10^5\ W/m^2 \cdot A)(0.0016\ \frac{m^2K}{W} \cdot A^{-1})$$

$$\boxed{T_s = 1104\,K}$$

thus, with the TBC, the Superalloy will be able to be maintained under 1200K and not exceed the maximum temperature